Mathematics Assessment 1 Total Marks: 40 Answer all questions.

Multiple Choice (20 marks)

1. Which of these vowels does NOT have a vertical axis of symmetry?
   - (a) A
   - (b) E
   - (c) I
   - (d) O
2. The practice range at a golf course charges $4.00 for a bucket of 40 golf balls. At this rate, how much will a bucket of 100 golf balls cost?
   - (a) $10.00
   - (b) $10.50
   - (c) $13.50
   - (d) $16.00
3. Solve 5 over 8 *m = 6.
   - (a) 3 and 5 over 9
   - (b) 3 and 3 over 4
   - (c) 5 and 1 over 3
   - (d) 9 and 3 over 5
4. Statement 1 — Every nonzero free abelian group has an infinite number of bases. Statement 2 — Every free abelian group of rank at least 2 has an infinite...
   - (a) True, True
   - (b) False, False
   - (c) True, False
   - (d) False, True
5. The polynomial which results from the expansion of __MATH_EXPR_0__ has degree __MATH_EXPR_1__. Find __MATH_EXPR_2__.
   - (a) -2
   - (b) -4
   - (c) -1
   - (d) 1
6. The curve defined by x^3 + xy - y^2 = 10 has a vertical tangent line when x =

(a) 0 or -1/3
(b) 1.037
(c) 2.074
(d) 2.096

7. Two numbers added together are 19. Their product is 70. What are the two numbers?

(a) 5, 14
(b) 7, 10
(c) 4, 15
(d) 3, 16

8. Which expression is equal to 720?

(a) 7x20
(b) 8x80
(c) 9x80
(d) 9x90

9. Statement 1 — If R is a ring and f(x) and g(x) are in R[x], then deg (f(x)+g(x)) = maximum (deg f(x), deg g(x)). Statement 2 — If F is a field then every...

(a) True, True
(b) False, False
(c) True, False
(d) False, True

10. Evaluate __MATH_EXPR_0__.

(a) 4
(b) 0.25
(c) -1
(d) 27

True / False (20 marks)

Answer True or False and justify briefly.

1. True or False: The correct answer to 'There are three real numbers __MATH_EXPR_0__ that are not in the domain of$__MATH_EXPR_1__$ What is the sum of those three...' is '-1.5'.

2. True or False: The correct answer to 'What is the value of p in 24 = 2p?' is 'p = 12'.

3. True or False: The correct answer to 'Three points are chosen randomly and independently on a circle. What is the probability that all three pairwise distances between the...' is '\{}frac{1}{12}'.

4. True or False: The correct answer to 'For how many positive integers k does the ordinary decimal representation of the integer k! end in exactly 99 zeros?' is 'Five'.

5. True or False: The correct answer to 'If A = {1, 2, 3} then relation S = {(1, 1), (2, 2)} is' is 'an equivalence relation'.

6. True or False: The correct answer to 'Which group of numbers is in order from least to greatest?' is '0.25 1.0 1.6'.

7. True or False: The correct answer to 'Suppose that for some __MATH_EXPR_0__ we have __MATH_EXPR_1__, __MATH_EXPR_2__. What is __MATH_EXPR_3__?' is '1'.

8. True or False: The correct answer to 'A subset H of a group (G,*) is a group if' is 'a,b in H=¿ a * bˆ-1 in H'.

9. True or False: The correct answer to 'If the parabola __MATH_EXPR_0__ and the line __MATH_EXPR_1__ intersect at only one point, what is the value of __MATH_EXPR_2__?' is '7'.

10. True or False: The correct answer to 'Stephanie flew 2,448 miles from Los Angeles to New York City. What is the number of miles Stephanie flew rounded to the nearest thousand?' is '2,000'.

End of Paper